

01.1 supplement • Sets, measurable maps and Bertrand's paradox

Data Analytics for AI • Executive M.Tech • IIT Jammu

Module 01 • Fundamentals of Probability and Random Signal Theory

Supplement to 01.1 • Probability through ticket triage

01.1 main materials

- **Notes:** Main lecture notes (PDF)
- **Deck:** Worked student deck (PDF)

Supplementary materials

- **Notes:** This rigorous supplement (PDF)
- **Deck:** Probability before Kolmogorov (PDF)
- **Deck:** Bertrand's paradox (PDF)
- **Interactive demonstration:** Sampling maps and chord length

This supplement gives the formal definitions, parametrizations and derivations supporting the main lecture. Its details are intended for supplementary study alongside the corresponding lecture topics. For the chord calculations, R is the fixed circle radius, T the normalized midpoint distance, and L the chord length, as in the Bertrand deck. The Billing indicator in the main notes remains X_B .

The main lecture motivates these constructions with questions about a hypothetical six-ticket routing audit, before introducing formal probability spaces. The proofs below do not depend on that application; they state the general hypotheses needed for the results.

Reference: Krishna Jagannathan, *Probability Foundations for Electrical Engineers*, 9 July 2025, Chapters 1, 4 and 11, particularly Definitions 68–70 for measurable functions, random variables and their laws. The calculations of the sampling maps are written out below.

1. Outcomes, events and mutual exclusivity

A sample space is required to satisfy two conditions. It must be **exhaustive**, so that at least one of its elements occurs in any realization of the experiment, and its elements must be **mutually exclusive**, so that at most one of them occurs. Together these state that exactly one $\omega \in \Omega$ occurs each time the experiment is performed.

Distinct outcomes correspond to disjoint singleton subsets. When these singletons belong to the event collection, they are singleton events:

$$\{\omega_1\} \cap \{\omega_2\} = \emptyset \quad \text{whenever } \omega_1 \neq \omega_2.$$

For countable Ω the singletons therefore form a countable disjoint partition, $\Omega = \bigcup_{\omega \in \Omega} \{\omega\}$ (a disjoint union).

Outcomes are mutually exclusive; events need not be. For a single roll of a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$, and the outcomes 2 and 4 cannot both occur. The events

$$A = \{2, 4, 6\} \text{ (even),} \quad B = \{4, 5, 6\} \text{ (greater than 3)}$$

can both occur, since $A \cap B = \{4, 6\} \neq \emptyset$.

Consequences for calculation

The two cases are handled by different rules, and the distinction above is what separates them.

Because singletons are disjoint, countable additivity applies directly to them, and for a countable event A whose singletons are measurable

$$P(A) = \sum_{\omega \in A} P(\{\omega\}).$$

For two events no such rule is available, since additivity requires disjointness. The correction term is the shared part:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Writing $P(A \cup B) = P(A) + P(B)$ for the die events above would count the outcomes 4 and 6 twice.

Two errors that this distinction prevents

A list of events is not a sample space. For one roll of a die, the pair "even" and "greater than 3" is not a sample space: both can occur in a single roll, so the second condition fails. They are events in the finer space $\{1, \dots, 6\}$.

A valid sample space need not have equally likely elements. For two independent fair tosses, the list "a head on the first toss", "a tail then a head", "two tails" is exhaustive and mutually exclusive, so it is a legitimate sample space. Its members carry probabilities 1/2, 1/4 and 1/4: they group the equally likely ordered outcomes as $\{HH, HT\}$, $\{TH\}$ and $\{TT\}$. This calculation also appears in the interpretations deck. Mutual exclusivity is a requirement on the sample space; it says nothing about the probabilities assigned to its elements, and counting the members is valid only when they happen to be equally probable.

The same reading in the Bertrand construction

The sampling procedures in Section 3 produce a geometric chord almost surely. After excluding the stated null cases, we may describe their outputs on the common space $\mathcal{C}_*$ of nondegenerate, nondiameter chords. In this output description, a chord is an elementary outcome. The three rotationally invariant models use the same chord space and event collection but different probability measures. Their original sampling outcomes can instead be angles, distances or points; Section 3 distinguishes these descriptions.

The event $E = \{c: L(c) > \sqrt{3} R\}$ is not an outcome. It is a set containing uncountably many chords, and it is assigned a probability as a set.

This last point also marks the limit of the summation rule above. In a continuous model each individual outcome has probability zero, so

$$\sum_{c \in E} P(\{c\}) = 0 \neq P(E),$$

and the probability of E cannot be recovered by adding the probabilities of the outcomes it contains. Countable additivity is additivity over countably many disjoint events, and E is not a countable union of its singletons. The rule $P(A) = \sum_{\omega \in A} P(\{\omega\})$ applies to countable events with measurable singletons; it does not extend to an uncountable event by countable additivity.

2. Random variables, inverse images and probability laws

On a probability space $(\Omega, \mathcal{F}, P)$, a real-valued **random variable** is a measurable function

$$X: \Omega \rightarrow \mathbb{R}.$$

Measurability means that $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \in \mathcal{B}(\mathbb{R})$. The Borel σ -algebra is generated by open intervals and includes the intervals, singleton sets and their countable combinations used here. For a real-valued function, it is sufficient to check $\{\omega: X(\omega) \leq x\} \in \mathcal{F}$ for every real x .

The distinction between a random variable and a set function

A random variable is a function **on outcomes**, not a real-valued set function on events. It induces the following map between collections of sets:

$$X^{-1}: \mathcal{B}(\mathbb{R}) \rightarrow \mathcal{F}, \quad B \mapsto \{\omega: X(\omega) \in B\}.$$

The probability measure and the distribution of X are real-valued **set functions**:

$$P: \mathcal{F} \rightarrow [0, 1], \quad P_X: \mathcal{B}(\mathbb{R}) \rightarrow [0, 1],$$

$$P_X(B) = P(X^{-1}(B)) = P(X \in B).$$

Thus a numerical question about X is first converted to a set of outcomes, and that event is assigned a probability. This is the set-based construction used in the Bertrand calculation below. It follows the distinction in Jagannathan, Chapter 11, Definitions 68–70.

Independence used in the examples

Events E, F are independent when $P(E \cap F) = P(E)P(F)$. Random variables U, V are independent when this identity holds for the events $\{U \in A\}$ and $\{V \in B\}$ for every pair of Borel sets A, B . For variables with a joint density on a product domain, $f_{U,V}(u, v) = f_U(u)f_V(v)$ almost everywhere implies independence: integrating over $A \times B$ separates into the product of the two marginal integrals. Independent fair tosses below stipulate the product measure, giving each ordered pair mass $(1/2)(1/2) = 1/4$.

Worked example: the head count

For two independent fair tosses, each outcome in $\Omega = \{HH, HT, TH, TT\}$ has probability $1/4$, and $\mathcal{F} = 2^\Omega$. Define $X(HH) = 2$, $X(HT) = X(TH) = 1$, and $X(TT) = 0$. Then

$$P_X(\{1\}) = P(\{HT, TH\}) = 1/2,$$

$$P_X([1, 2]) = P(\{HH, HT, TH\}) = 3/4.$$

The distribution assigns probabilities $1/4, 1/2, 1/4$ to $0, 1, 2$. It is obtained through inverse images, not by assuming that the three numerical values are equally likely.

Worked example: an indicator

For an event $A \in \mathcal{F}$, its indicator is the random variable

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Its inverse images satisfy $\mathbf{1}_A^{-1}(\{1\}) = A$ and $\mathbf{1}_A^{-1}(\{0\}) = A^c$. Consequently, $P(\mathbf{1}_A = 1) = P(A)$.

For the parity model in the event-collection lab, take $\mathcal{G} = \{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, \Omega\}$. The indicator of an even face is measurable. The numerical face-value function is not measurable relative to that σ -algebra, since the inverse image of $\{6\}$ is the unavailable subset $\{6\}$.

3. Bertrand's paradox: sampling maps and induced distributions

This section develops the Bertrand question posed in Section 3 of the main notes. What is the probability that a randomly generated chord is longer than the side of an inscribed equilateral triangle? We first calculate three answers to this same plain-English question. Only then do we use the Kolmogorov triplet to identify precisely which probability problem each calculation answers.

The paradox: one question, three answers

For the unit circle, a chord at distance r from the centre has length

$$L = 2\sqrt{1 - r^2}.$$

Thus $L > \sqrt{3}$ exactly when $r < 1/2$, or equivalently when the chord intersects the interior of the concentric disk of radius $1/2$.

Consider three plausible readings of "choose a chord at random."

1. **Choose its angle uniformly at a fixed endpoint.** Fix $a = (1, 0)$ and measure an angle $\theta \in (0, \pi)$ from the upward tangent towards the disk. The resulting chord has length $2\sin \theta$. It is longer than $\sqrt{3}$ when $\pi/3 < \theta < 2\pi/3$, an interval occupying one third of the available angular range. The answer is $1/3$.
2. **Choose its distance from the centre uniformly.** Fix the midpoint direction and choose its distance r uniformly between 0 and 1 . Draw the chord perpendicular to that radius. The favorable distances satisfy $r < 1/2$, half the available range. The answer is $1/2$.

3. **Choose its midpoint uniformly in disk area.** Draw the chord perpendicular to the radius through the chosen midpoint. It is long enough exactly when the midpoint lies inside the concentric disk of radius $1/2$. That disk occupies one quarter of the total area. The answer is $1/4$. The centre and boundary are zero-area exceptions handled below.

We have asked the same question in words and obtained

$$\frac{1}{3}, \quad \frac{1}{2}, \quad \frac{1}{4}.$$

Each calculation is correct for its construction. The phrase “a random chord” leaves the sampling rule unspecified: uniform angle, distance and area are different assumptions. We now use the Kolmogorov triplet to resolve the apparent contradiction.

Making the experiments precise with the Kolmogorov triplet

A probability model is a triplet $(\Omega, \mathcal{F}, P)$: Ω specifies the outcomes, $\mathcal{F}$ specifies the measurable events, and P assigns their probabilities. We now identify these three components for each calculation. We distinguish the common geometric question from the event that represents it in each sampling space.

Experiment I: a uniform angle at a fixed endpoint. Fix the endpoint $a = (1, 0)$. From the upward tangent at a , measure $\theta \in (0, \pi)$ towards the interior of the disk. The ray has direction $(-\sin \theta, \cos \theta)$; its second intersection with the circle defines the chord. Substituting $a + \ell(-\sin \theta, \cos \theta)$ into the circle equation gives $\ell(\ell - 2\sin \theta) = 0$, so its length is $L = 2\sin \theta$.

Choose θ uniformly. The triplet and measure are

$$\Omega_1 = (0, \pi), \quad \mathcal{F}_1 = \mathcal{B}((0, \pi)), \quad P_1(B) = \frac{1}{\pi} \int_B d\theta \quad (B \in \mathcal{F}_1).$$

Here $\mathcal{B}$ is the Borel σ -algebra, generated by the open subsets of the stated domain. The favorable event and probability are

$$E_1 = (\pi/3, 2\pi/3), \quad P_1(E_1) = \frac{2\pi/3 - \pi/3}{\pi} = \frac{1}{3}.$$

The endpoints $\theta = 0, \pi$ would give zero-length chords and have been excluded. The diameter at $\theta = \pi/2$ has probability zero and will also be omitted when we introduce the common chord chart.

Experiment II: a uniform distance with a fixed direction. Choose $r \in (0, 1)$ uniformly, set the midpoint to $(r, 0)$, and draw the vertical chord through it. Specifying the midpoint direction makes the chord unique; distance alone determines only its length.

$$\Omega_2 = (0, 1), \quad \mathcal{F}_2 = \mathcal{B}((0, 1)), \quad P_2(B) = \int_B dr \quad (B \in \mathcal{F}_2).$$

The favorable event and probability are

$$E_2 = (0, 1/2), \quad P_2(E_2) = \int_0^{1/2} dr = \frac{1}{2}.$$

Experiment III: a uniform midpoint in area. Choose a point $M = (x, y)$ uniformly in the disk and draw the chord perpendicular to OM through M . To make this construction unique and nondegenerate everywhere on its domain, exclude the centre and boundary, both of which have area zero. Put

$$D^* = \{(x, y): 0 < x^2 + y^2 < 1\}.$$

Then

$$\Omega_3 = D^*, \quad \mathcal{F}_3 = \mathcal{B}(D^*), \quad P_3(B) = \frac{1}{\pi} \iint_B dx \, dy \quad (B \in \mathcal{F}_3).$$

The favorable event is the punctured inner disk, so

$$E_3 = \{(x, y): 0 < x^2 + y^2 < 1/4\}, \quad P_3(E_3) = \frac{\pi(1/2)^2}{\pi} = \frac{1}{4}.$$

Deleting the centre does not change either area. Using the closed disk with an explicit convention for the exceptional cases gives the same probabilities.

Experiment	Sample space	Measure on Borel events	Favorable event	Probability
Fixed endpoint, uniform angle	$(0, \pi)$	$d\theta/\pi$	$\pi/3 < \theta < 2\pi/3$	1/3
Fixed direction, uniform distance	$(0, 1)$	dr	$r < 1/2$	1/2
Uniform midpoint area	D^*	$dx \, dy/\pi$	$x^2 + y^2 < 1/4$	1/4

The notation for the measures in the table means integration over an event, as defined above. **With the probability triplets specified, the apparent contradiction is resolved:** the three answers belong to three different probability models.

Advanced: measures on a common chord space

The triplets above complete the resolution. The remaining calculations study sampling maps and induced measures on geometric chords. They are not needed to resolve the paradox.

We consider three rotationally invariant models independently:

- **Model A:** choose two independent uniform endpoints on the circumference.
- **Model B:** choose a uniform radial distance and an independent uniform midpoint direction.
- **Model C:** choose a midpoint uniformly in disk area.

Their connection to Experiments I–III is Exercise 8. In particular, equal chord-length distributions need not mean equal measures on geometric chords. A common sample space does not require identical supports.

The endpoint, distance and area constructions are also discussed in Larson and Odoni, *Urban Operations Research*, §3.3.2. We now specify the common measurable space and derive each model's sampling law.

The event and a common chord space

Fix a circle of radius $R > 0$. The side of an inscribed equilateral triangle has length $\sqrt{3} R$. For a chord c , define the real-valued random variables

$$D(c) = \text{distance from the centre to } c, \quad T(c) = D(c)/R, \quad L(c) = \text{length of } c.$$

The geometry gives

$$L(c) = 2R\sqrt{1 - T(c)^2}, \quad E = \{c: L(c) > \sqrt{3} R\} = \{c: T(c) < 1/2\}.$$

We compare the probability of this **same event** under three probability measures on a common chord space.

For a one-to-one parametrization, let $\mathcal{C}_*$ be the set of unordered, nondegenerate chords that are not diameters. Coincident-endpoint chords and diameters have probability zero under all three rules below. Excluding them makes the inverse formulas unambiguous without changing any calculated probability. They may instead be retained in an enlarged space with total probability zero.

A chord parametrization and its inverse

Let $0 < t < 1$ and $0 \leq \phi < 2\pi$. Write

$$n(\phi) = (\cos \phi, \sin \phi), \quad n_\perp(\phi) = (-\sin \phi, \cos \phi).$$

Define $C(\phi, t)$ to be the chord with unordered endpoints

$$z_\pm = Rt n(\phi) \pm R\sqrt{1 - t^2} n_\perp(\phi).$$

Its midpoint is $m = Rt n(\phi)$, so the inverse is

$$C^{-1}(c) = \left(\text{Arg}(m), \frac{\|m\|}{R} \right), \quad m = \frac{z_+ + z_-}{2}.$$

Here Arg is chosen in $[0, 2\pi)$. The midpoint is nonzero because diameters were excluded. Hence $C: [0, 2\pi) \times (0, 1) \rightarrow \mathcal{C}_*$ is a bijection. Write $K = [0, 2\pi) \times (0, 1)$ and define the event collection explicitly by

$$\mathcal{F}_{\mathcal{C}_*} = \{B \subseteq \mathcal{C}_*: C^{-1}(B) \in \mathcal{B}(K)\}.$$

This makes C and its inverse measurable. The three chord-space triplets are

$$(\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_A), \quad (\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_B), \quad (\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_C).$$

Their sample space, event collection and long-chord event E are identical; their measures differ. The sampling maps below define these measures.

The length function L is **not** a bijection on chords: all orientations with the same t have the same length. The inverse image $L^{-1}(B)$ is still defined for every Borel set of lengths B .

A common source of uniform samples

Use the probability space

$$\Omega_0 = (0, 1) \times [0, 1), \quad \mathcal{F}_0 = \mathcal{B}(\Omega_0), \quad Q = \text{uniform area measure on } \Omega_0.$$

The coordinate random variables $U(u, v) = u$ and $V(u, v) = v$ are independent uniform variables. Endpoints of these intervals have measure zero. For each mechanism $j \in \{A, B, C\}$, define a sampling map

$$\Psi_j(u, v) = C(2\pi v, q_j(u)).$$

Each q_j given below is a continuous bijection of $(0, 1)$. Since $C^{-1} \circ \Psi_j(u, v) = (2\pi v, q_j(u))$ is continuous into K , the definition of $\mathcal{F}_{\mathcal{C}^*}$ makes Ψ_j measurable. It is also a bijection onto $\mathcal{C}^*$. The event E is measurable because $C^{-1}(E) = [0, 2\pi) \times (0, 1/2)$.

The induced probability measure on chord events is

$$P_j(F) = Q(\Psi_j^{-1}(F)), \quad F \in \mathcal{F}_{\mathcal{C}^*}.$$

The chord-length random variable on the uniform source space is $L_j = L \circ \Psi_j$. We now specify each q_j , its inverse and the distribution it produces.

A. Two independent uniform endpoints

The original procedure selects independent angles $A_1, A_2 \sim \text{Unif}[0, 2\pi)$ and joins the points $R(\cos A_i, \sin A_i)$. The joint angle density is $1/(2\pi)^2$.

Let $\delta \in (-\pi, \pi]$ be the signed principal difference $A_2 - A_1$ modulo 2π , and let $H = |\delta|/2$. In the deck's notation, $\alpha = |\delta| = 2H$. The cases $\delta = 0$ (coincident endpoints) and $|\delta| = \pi$ (diameters) have probability zero and are excluded for the chord chart. The map $(a_1, a_2) \mapsto (a_1, \delta)$ is a bijection apart from this null boundary and the angular wrap. For each a_1 , the second coordinate is a translation modulo 2π , so the map preserves area. Hence (A_1, δ) has constant density $1/(4\pi^2)$ on the rectangle.

Now define $\Phi = (A_1 + \delta/2) \bmod 2\pi$. The map $(a_1, \delta) \mapsto (\Phi, \delta)$ is also one-to-one on the chosen half-open angular ranges and preserves area: for fixed δ it is a translation in the first coordinate. Thus (Φ, δ) has the same constant density, which factors into uniform marginal densities. Applying $H = |\delta|/2$ gives the joint density

$$f_{\Phi, H}(\phi, h) = \frac{1}{2\pi} \frac{2}{\pi}, \quad 0 \leq \phi < 2\pi, \quad 0 < h < \pi/2.$$

Therefore Φ and H are independent, Φ is uniform, and $H \sim \text{Unif}(0, \pi/2)$. The midpoint is in direction Φ , since the midpoint vector is proportional to $\cos(H) n(\Phi)$ and $\cos(H) > 0$ almost surely.

The ordered-endpoint map is two-to-one because interchanging endpoints leaves the chord unchanged. Its inverse therefore recovers an **unordered pair**, not a unique ordered pair. From (ϕ, t) , put $h = \arccos(t)$; the unordered endpoint angles are

$$\{(\phi - h) \bmod 2\pi, (\phi + h) \bmod 2\pi\}.$$

Thus an equivalent, bijective sampling parametrization of the unordered chords uses $U = 2H/\pi$ and $V = \Phi/(2\pi)$:

$$q_A(u) = \cos(\pi u/2), \quad \Psi_A^{-1}(c) = \left(\frac{2}{\pi} \arccos(T(c)), \frac{\text{Arg}(m)}{2\pi} \right).$$

The length random variable and the inverse image of the event are

$$L_A(u, v) = 2R \sin(\pi u/2), \quad \Psi_A^{-1}(E) = \{(u, v) : u > 2/3\}.$$

Consequently,

$$P_A(E) = Q(U > 2/3) = 1/3.$$

For $0 < t < 1$, the induced distribution of normalized midpoint distance is

$$F_{T,A}(t) = 1 - \frac{2}{\pi} \arccos(t) = \frac{2}{\pi} \arcsin(t), \quad f_{T,A}(t) = \frac{2}{\pi \sqrt{1-t^2}}.$$

It is not uniform in radial distance.

B. Uniform radial distance

Choose $D \sim \text{Unif}(0, R)$ and an independent direction $\Phi \sim \text{Unif}[0, 2\pi)$. Construct the perpendicular chord through D $n(\Phi)$. Equivalently,

$$q_B(u) = u, \quad \Psi_B^{-1}(c) = \left(T(c), \frac{\text{Arg}(m)}{2\pi} \right).$$

The original (D, Φ) joint density is $1/(2\pi R)$ on $(0, R) \times [0, 2\pi)$. In normalized coordinates, T has density $f_{T,B}(t) = 1$ and distribution function $F_{T,B}(t) = t$ for $0 < t < 1$.

$$L_B(u, v) = 2R\sqrt{1-u^2}, \quad \Psi_B^{-1}(E) = \{(u, v): u < 1/2\}.$$

Therefore

$$P_B(E) = Q(U < 1/2) = 1/2.$$

C. Uniform midpoint in disk area

Choose a midpoint $M = (M_1, M_2)$ with Cartesian density $1/(\pi R^2)$ inside the disk. In polar coordinates $M = Rt$ $n(\phi)$, the area element is $R^2 t$ dt $d\phi$. Hence

$$f_{T,\Phi}(t, \phi) = \frac{t}{\pi} = 2t \cdot \frac{1}{2\pi}, \quad 0 < t < 1.$$

Thus T and Φ are independent, Φ is uniform, and T has density $2t$ and distribution function $F_{T,C}(t) = t^2$ for $0 < t < 1$. A uniform draw U is transformed by

$$q_C(u) = \sqrt{u}, \quad \Psi_C^{-1}(c) = \left(T(c)^2, \frac{\text{Arg}(m)}{2\pi} \right).$$

The length random variable and event preimage are

$$L_C(u, v) = 2R\sqrt{1-u}, \quad \Psi_C^{-1}(E) = \{(u, v): u < 1/4\}.$$

Therefore

$$P_C(E) = Q(U < 1/4) = 1/4.$$

The area ratio $\pi(R/2)^2/(\pi R^2)$ gives the same result.

Explicit comparison on the same coordinates

In every model, Φ is independent of T and uniform on $[0, 2\pi)$. The density with respect to $d\phi dt$ is $f_{T,j}(t)/(2\pi)$. Thus, for every $B \in \mathcal{F}_{\mathcal{C}_*}$,

$$P_j(B) = \iint_{C^{-1}(B)} \frac{f_{T,j}(t)}{2\pi} d\phi dt.$$

Equivalently, these densities define three measures on the same coordinate space $(K, \mathcal{B}(K))$.

Model	$T = q_j(U)$	Inverse $U = q_j^{-1}(T)$	Density $f_{T,j}(t)$, $0 < t < 1$	$P_j(E)$
Uniform endpoints	$\cos(\pi U/2)$	$(2/\pi)\arccos(T)$	$2/(\pi\sqrt{1-t^2})$	1/3
Uniform radial distance	U	T	1	1/2
Uniform midpoint area	$\sqrt{U}$	T^2	$2t$	1/4

For each row,

$$P_j(E) = \int_0^{1/2} f_{T,j}(t) dt = Q(\Psi_j^{-1}(E)).$$

There is no equality between $P_A(E)$, $P_B(E)$ and $P_C(E)$ to be contradicted: the measures are different. By contrast, on the common uniform source space $(\Omega_0, \mathcal{F}_0, Q)$, the measure Q stays fixed while the sampling maps change. The event preimages $\Psi_j^{-1}(E) = \{(u, v): L_j(u, v) > \sqrt{3}/R\}$ are different sets, with Q -measures 1/3, 1/2, and 1/4. Within a fixed model, a correct reparametrization preserves event probability.

A reparametrization within one fixed model

Retain Model B and set $W = T^2$. Since T is uniform on $(0, 1)$,

$$P_B(W \leq w) = P_B(T \leq \sqrt{w}) = \sqrt{w}, \quad f_{W,B}(w) = \frac{1}{2\sqrt{w}}, \quad 0 < w < 1.$$

The inverse transformation is $T = \sqrt{W}$. The same event is now $W < 1/4$, and

$$P_B(E) = \int_0^{1/4} \frac{dw}{2\sqrt{w}} = \frac{1}{2}.$$

If W were instead declared uniform, its density would be changed to 1. That would give Model C and probability 1/4. This would be a change of probability model, not merely a change of coordinates.

The distribution of chord length (additional calculation)

For $0 \leq x \leq 2R$, put $z = x/(2R)$. Because $L = 2R\sqrt{1 - T^2}$ is decreasing in T ,

$$P_j(L \leq x) = P_j(T \geq \sqrt{1 - z^2}).$$

The three chord-length distribution functions are

$$F_{L,A}(x) = \frac{2}{\pi} \arcsin(z), \quad F_{L,B}(x) = 1 - \sqrt{1 - z^2}, \quad F_{L,C}(x) = z^2.$$

They are 0 for $x < 0$ and 1 for $x > 2R$. Substituting $x = \sqrt{3} R$ and taking $1 - F_{L,j}(x)$ gives $1/3, 1/2, 1/4$. These continuous laws assign zero probability to the equality at the threshold.

The set-based definitions follow Jagannathan, Chapters 1, 4 and 11. The comparison in common distance and angular coordinates is also developed in *Urban Operations Research*, §3.3.2. The explicit sampling maps and inverse calculations above are provided here in full.

4. The induced law is a probability measure

Let $X: (\Omega, \mathcal{F}, P) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable and define $P_X(B) = P(X^{-1}(B))$. Nonnegativity follows from that of P , and

$$P_X(\mathbb{R}) = P(X^{-1}(\mathbb{R})) = P(\Omega) = 1.$$

If $B_1, B_2, \dots$ are pairwise disjoint Borel sets, their inverse images are pairwise disjoint events, and inverse images preserve their union. Therefore

$$P_X\left(\bigcup_{n=1}^{\infty} B_n\right) = P\left(\bigcup_{n=1}^{\infty} X^{-1}(B_n)\right) = \sum_{n=1}^{\infty} P(X^{-1}(B_n)) = \sum_{n=1}^{\infty} P_X(B_n).$$

This proves that P_X is a probability measure. The same proof applies to the sampling maps Ψ_j and the induced measures P_j on the chord space. Injectivity is unnecessary for this construction: inverse images exist for all functions.

Distinct meanings of the notation f^{-1}

When $f: S \rightarrow Y$ is a bijection, $f^{-1}(y)$ denotes an element of S , the value of the inverse function at y . For any subset $B \subseteq Y$, the notation $f^{-1}(B)$ denotes a subset of S , the inverse image. The latter is defined even when f is not one-to-one. In the Bertrand construction, C and each Ψ_j are bijections on the stated restricted domains, whereas the chord-length function L is not one-to-one.

5. Why a rotation-invariant probability cannot measure every subset

The main notes use Borel sets for the continuous model. Why not simply use every subset, as in the finite audit? Under the axiom of choice, the following construction shows that a normalized, countably additive, rotation-invariant probability on a circle cannot be defined on all its subsets.

The setting

Let $\Omega = [0, 1)$ and define addition modulo 1,

$$x \oplus y = \begin{cases} x + y, & x + y < 1, \\ x + y - 1, & x + y \geq 1, \end{cases}$$

with $\ominus$ the corresponding subtraction. Under $\oplus$ the set $[0, 1)$ is an abelian group; identifying 0 with 1 makes it a circle, and $x \mapsto x \oplus t$ is a rotation. For $A \subseteq [0, 1)$ write $A \oplus t = \{a \oplus t : a \in A\}$.

Suppose P assigns a nonnegative real number to **every** subset of $[0, 1)$ and satisfies

1. $P([0, 1)) = 1$;
2. countable additivity;
3. rotation invariance: $P(A \oplus t) = P(A)$ for all A and all $t \in [0, 1)$.

The construction

Let $\mathbb{Q}_0 = \mathbb{Q} \cap [0, 1)$, a countable subgroup of $([0, 1), \oplus)$. Define $x \sim y$ when $x \ominus y \in \mathbb{Q}_0$. This is an equivalence relation, and its classes are the cosets $x \oplus \mathbb{Q}_0$.

By the axiom of choice, let $V \subseteq [0, 1)$ contain exactly one element of each class.

Claim 1: the sets $V \oplus q$, for $q \in \mathbb{Q}_0$, are pairwise disjoint. Suppose $v_1 \oplus q_1 = v_2 \oplus q_2$ with $v_1, v_2 \in V$. Then $v_1 \ominus v_2 = q_2 \ominus q_1 \in \mathbb{Q}_0$, so $v_1 \sim v_2$. Since V has one element per class, $v_1 = v_2$, and therefore $q_1 = q_2$.

Claim 2: these sets cover $[0, 1)$. Given $x \in [0, 1)$, let $v \in V$ be the representative of the class of x . Then $q = x \ominus v \in \mathbb{Q}_0$ and $x = v \oplus q \in V \oplus q$.

So $[0, 1)$ is the union of countably many pairwise disjoint sets, each a rotation of V .

The contradiction

Write $c = P(V)$. Rotation invariance gives $P(V \oplus q) = c$ for every q , and countable additivity applied to the countable disjoint family of Claim 1, whose union is $[0, 1)$ by Claim 2, gives

$$1 = P([0, 1)) = \sum_{q \in \mathbb{Q}_0} P(V \oplus q) = \sum_{q \in \mathbb{Q}_0} c.$$

Since P is nonnegative, $c \geq 0$. If $c = 0$ the right-hand side is 0; if $c > 0$ the sum diverges. Neither equals 1, so no such P exists.

Moreover, if V were Lebesgue measurable, each $V \oplus q$ would be measurable with the same Lebesgue measure: addition modulo 1 is an ordinary translation on each of two pieces. The same countable sum would then contradict the Lebesgue measure of $[0, 1)$ being 1. Thus V is not Lebesgue measurable. This is the Vitali construction on the circle.

Scope of the conclusion

The contradiction uses both countable additivity and rotation invariance. Finite additivity alone does not justify the countably infinite sum above. Removing invariance changes the question: for example, the point mass $P(A) = \mathbf{1}_{\{0 \in A\}}$ is a countably additive probability on every subset of $[0, 1)$, but it is not rotation invariant.

The selection of V uses the axiom of choice; the argument does not give a membership rule for V . No restriction proved here prevents assigning probabilities to the Borel events used in this course. Intervals, disks, angular sectors and the sampling-map preimages in Section 3 are measurable. The construction explains a boundary of the all-subsets approach; it is not needed to carry out the practical calculations in the main lesson.

6. Exercises on the supplement

1. Verify both $C^{-1}(C(\phi, t)) = (\phi, t)$ and $C(C^{-1}(c)) = c$ on the stated domains. Explain why the restriction on t is needed.
2. Verify both inverse identities for $q_A(u) = \cos(\pi u/2)$ on $(0, 1)$. Repeat for $q_C(u) = \sqrt{u}$.
3. Integrate each density in the comparison table over $(0, 1)$ and over $(0, 1/2)$.
4. Derive the three chord-length distribution functions from the distance distribution functions.
5. Prove that the inverse image of a union is the union of the inverse images. Use this to explain why the induced law is countably additive.
6. For one roll of a die, decide whether each of the following is a sample space, giving a reason: (a) $\{1, \dots, 6\}$; (b) $\{\text{even, odd}\}$; (c) $\{\text{even, greater than 3}\}$; (d) $\{\text{less than 3, equal to 3, greater than 3}\}$.
7. In the Vitali construction, identify the step that fails if $\mathbb{Q}_0$ is replaced by a finite subgroup of $([0, 1), \oplus)$, and state what the argument then yields.
8. Connect the elementary experiments to Models A–C. (a) For Experiment I, find the second endpoint from $a = (1, 0)$ and angle θ . Then randomize the first endpoint uniformly, independently of θ , and prove that the two endpoint angles are independent and uniform. Show that the length law is unchanged. (b) Give Experiment II an independent uniform midpoint direction. Explain why its length law is unchanged. (c) Identify the relation between Experiment III and Model C. Finally, compare the probability of containing a in Experiment I and Model A. Does an equal length law imply an equal chord measure?
9. The density $f_{T,A}(t) = 2/(\pi\sqrt{1-t^2})$ is unbounded as $t \uparrow 1$. Verify that it is integrable and find $P_A(T > 1 - \varepsilon)$ for $0 < \varepsilon < 1$. What happens as $\varepsilon \downarrow 0$?
10. In Model B, write the event $T < 1/3$ in terms of $W = T^2$. Calculate its probability using the distribution function of W .

Solutions

1. If $c = C(\phi, t)$, its midpoint is $Rt \, n(\phi)$, so $C^{-1}(c) = (\phi, t)$. Conversely, for $c \in \mathcal{C}_*$, the perpendicular from the circle centre meets the chord at its midpoint and the two endpoints lie a distance $R\sqrt{1-t^2}$ on either side. These are exactly the endpoints in the definition of $C(\phi, t)$, so $C(C^{-1}(c)) = c$. At $t = 0$, the midpoint is zero and its argument is undefined; at $t = 1$, the endpoints coincide. Both cases were excluded and have probability zero under the three laws.
2. On the stated domains, $q_A^{-1}(t) = (2/\pi)\arccos(t)$. Because $\pi u/2 \in (0, \pi/2)$ lies in the principal range of $\arccos$, $\arccos(\cos(\pi u/2)) = \pi u/2$. Hence $q_A^{-1}(q_A(u)) = u$. Conversely, $\cos(\arccos(t)) = t$, so $q_A(q_A^{-1}(t)) = t$. For Model C, $q_C^{-1}(t) = t^2$, so $(\sqrt{u})^2 = u$ and $\sqrt{t^2} = t$ because $t > 0$.
3. For A, an antiderivative is $(2/\pi)\arcsin(t)$, giving total mass 1 and mass 1/3 below 1/2. For B, the corresponding values are $[t]_0^1 = 1$ and $[t]_0^{1/2} = 1/2$. For C, they are $[t^2]_0^1 = 1$ and $[t^2]_0^{1/2} = 1/4$.
4. Substitute $b = \sqrt{1 - (x/(2R))^2}$ into $1 - F_{T,j}(b)$ for $0 \leq x \leq 2R$. Use $\arccos(b) = \arcsin(x/(2R))$ in Model A. The variables have continuous laws, so the endpoint inequalities do not change the probabilities.
5. An outcome belongs to $f^{-1}(\bigcup_n B_n)$ exactly when its image belongs to at least one B_n , equivalently when it belongs to at least one $f^{-1}(B_n)$. The proof in Section 4 then applies.
6. For case (a), yes: exactly one face shows. (b) Yes: the two are exhaustive and cannot both occur. (c) No: both occur when the face is 4 or 6, so the elements are not mutually exclusive. (d) Yes: the three are exhaustive and pairwise disjoint. For a fair die the two elements of (b) have probability 1/2 each, whereas those of (d) have probabilities 2/6, 1/6 and 3/6. Both are sample spaces; equal probabilities are not part of the requirement.
7. Claim 1 and Claim 2 still hold, so $[0, 1)$ is a finite disjoint union of n rotations of V and $1 = nc$, giving $c = 1/n$. The final dichotomy, that the sum is either zero or divergent, no longer applies. The modified argument only requires $P(V) = 1/n$; it does not establish the existence of a rotation-invariant probability on all subsets. The original countably infinite subgroup argument still rules out such a probability.
8. **(a)** The second endpoint is $a + 2\sin \theta(-\sin \theta, \cos \theta) = (\cos(2\theta), \sin(2\theta))$. Thus its angle 2θ is uniform on $(0, 2\pi)$. After rotating by an independent uniform first-endpoint angle A_1 , the second angle is $A_2 = (A_1 + 2\theta) \bmod 2\pi$. Conditional on any $A_1 = a_1$, A_2 is uniform, with a law independent of a_1 ; hence A_1, A_2 are independent uniforms. Rotation preserves length, so the law of $2\sin \theta$ is unchanged. **(b)** Rotation of the midpoint direction preserves r and therefore $L = 2\sqrt{1-r^2}$; this gives Model B. **(c)** Experiment III already is Model C for $R = 1$. The probability of containing a is 1 in Experiment I and 0 in Model A: a chord contains a given circumference point only as an endpoint, and each uniform endpoint equals that point with probability zero. Equal length laws therefore need not give equal chord measures.
9. The substitution $t = \sin y$ gives $\int_0^1 2/(\pi\sqrt{1-t^2}) dt = (2/\pi) \int_0^{\pi/2} dy = 1$. Also $P_A(T > 1 - \varepsilon) = 1 - (2/\pi)\arcsin(1 - \varepsilon) = (2/\pi)\arccos(1 - \varepsilon) \rightarrow 0$. An unbounded density need not have infinite mass or an atom at the endpoint.
10. The event is $W < 1/9$. Its probability is $P(W < 1/9) = \sqrt{1/9} = 1/3$. Assigning probability 1/9 would incorrectly treat W as uniform under Model B.